

Sheila Kahwai

Backend/ML Engineer

Experienced software engineer specializing in backend systems, with expertise in API development and Third-Party API integration, machine learning, relational databases, cloud computing, CI/CD, containerization, and test-driven development. Bachelor's degree in Computer Science and a proven track record of impactful contributions to open-source and startup projects.

Contact

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Email

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GitHub

<https://github.com/Sheila-nk>

LinkedIn

<https://www.linkedin.com/in/sheila-kahwai>

Portfolio

<https://sheila-nk.github.io/>

Skills

- Python
- AWS & GCP
- Flask & FastAPI
- PostgreSQL
- Docker
- Tensorflow

Education

May 2022 to Nov 2022

The African Leadership(ALX) Software Engineering Programme

Apr 2019 to Aug 2019

The Tony Elumelu Foundation Entrepreneurship Programme

Sept 2014 to Nov 2018

Bsc. Computer Science

Jomo Kenyatta University of Agriculture and Technology

References

Available on Request

Experience

Jan 2023 - Present

Shelfy Africa | Remote, Kenya

Backend/ML Engineer

- Spearheaded the design and development of a robust RestfulAPI with Flask, ensuring seamless integration and optimal functionality.
- Developed and automated a model training pipeline, optimizing data preprocessing, model training, and evaluation processes, resulting in an efficient and streamlined machine learning workflow.
- Implemented an image retrieval and recommendation system using TensorFlow, integrating deep learning models to enhance visual search capabilities and improve personalized recommendations.
- Engineered and optimized a PostgreSQL database, ensuring data integrity, security, and high performance.
- Orchestrated application containerization via Docker, streamlining deployment and enhancing scalability.
- Managed configuration and maintenance of AWS cloud storage (S3), server (EC2), and database solutions (RDS), ensuring seamless operations, and robust infrastructure.

July 2023 - September 2023

Quansight Labs | Remote

Open Source Engineering Intern

- Developed a robust pytest plugin to enhance doctesting capabilities for PyData libraries, improving testing efficiency and accuracy.
- Integrated the pytest plugin into the Scipy repository, enhancing the doctesting experience for Scipy users and contributors.
- Automated the collection and execution of doctests across Scipy's module library using Pytest, streamlining the testing process and reducing manual intervention.
- Identified and resolved doctest failures across 7 Scipy modules by leveraging the pytest plugin, contributing to the library's overall reliability, quality, and accuracy by 15%.
- Crafted unit tests by utilizing the pytest module to verify the accuracy and reliability of the plugin.

Dec 2022 - March 2023

Project JupyterHub | Remote

Outreachy Software Engineering Intern

- Developed a reusable pytest plugin to improve the testing infrastructure of JupyterHub and its components.
- Led the project's packaging infrastructure setup and configuration using Hatchling and Setuptools.
- Configured and automated the project's document generation and hosting on ReadTheDocs using Sphinx.
- Implemented CI/CD practices for the Pytest Plugin using GitHub Workflows to reduce build and test times and improve collaboration within the JupyterHub community.
- Published the reusable JupyterHub Pytest Plugin to PyPI, contributing to its accessibility and usability.